

# RITHVIK PRANAO NAGARAJ

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## EDUCATION

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| <b>George Mason University</b><br>Master of Science in Computer Science                                       | Fairfax, VA<br>May 2026       |
| <b>Kumaraguru College of Technology</b><br>Bachelor of Technology in Artificial Intelligence and Data Science | Coimbatore, India<br>May 2024 |

## TECHNICAL SKILLS

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|------------------------------------|---------------------------------------------------------------------------|
| <b>Languages:</b>                  | Python, C#, R, JavaScript, SQL, HTML/CSS,                                 |
| <b>Systems &amp; DevOps:</b>       | AWS, Docker, CI/CD, Git                                                   |
| <b>Frontend &amp; Backend:</b>     | React, Angular, Node.js, FastAPI, RESTful APIs, Microservices, WebSockets |
| <b>Machine Learning/Libraries:</b> | TensorFlow, PyTorch(CUDA), Scikit-learn, LangChain, RAG                   |
| <b>Databases:</b>                  | PostgreSQL, MySQL, SQLite, DynamoDB, ChromaDB                             |

## EXPERIENCE

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| <b>George Mason University</b><br><i>Research Assistant</i> | Fairfax, VA<br>Jan 2025 – May 2025 |
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- Built Python automation tooling (pandas, BeautifulSoup, requests) to scrape and process 10,000+ cybersecurity posts.
- Cleaned CVE IDs, titles, and dates using regex and built ETL pipelines to output structured JSON datasets.
- Built an end-to-end RAG pipeline with LangChain and PGVector, embedding 10,000+ cybersecurity posts using all-MiniLM-L6-v2 and feeding retrieved context directly into a security LLM to reduce hallucinated output.
- Chunked and indexed unstructured security documents into a vector store, tuning embedding batch sizes and similarity thresholds until retrieval precision held up across varied document types.

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| <b>George Mason University</b><br><i>Teaching Assistant (Applied Deep Learning)</i> | Fairfax, VA<br>Aug 2025 – Dec 2025 |
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- Graded TensorFlow assignments for Applied Deep Learning, providing fast feedback to help students improve.
- Debugged student model implementations and explained core architecture concepts across CNNs, RNNs, and Transformers.

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| <b>Tradala Solution</b><br><i>Software Engineer Intern</i> | Chennai, India<br>Aug 2023 – Feb 2024 |
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- Refactored 2K+ lines of Node.js logic into a Unity C# SDK, reducing partner system integration latency by 30%.
- Integrated Ethereum smart contracts via Nethereum to manage high-throughput, secure transactional states natively.
- Designed and prototyped an asynchronous LangChain RAG pipeline in Python, ingesting 500+ pages of unstructured product specs into ChromaDB with retrieval filters validating JSON payloads, cutting manual spec lookup time by 30

## PROJECTS

### Flowboard – Real-Time Full-Stack Project Management Platform | *Angular, FastAPI, PostgreSQL, WebSockets*

- Architected a full-stack Kanban tool using Signals/RxJS and FastAPI with server-enforced role-based access control.
- Implemented real-time sync via native WebSockets, broadcasting live task and comment updates to active project members.
- Built a drag-and-drop (Angular CDK) with a database-backed invite/accept workflow.
- Added automated test coverage (pytest, Jasmine/Karma) and deployed to production with JWT authentication.

### Latent Space Model Optimization (Diffusion vs. GANs) | *PyTorch, Stable Diffusion VAE, MLOps*

- Implemented a 115M parameter Latent Diffusion Model based on the ADM architecture, training in compressed  $32 \times 32 \times 4$  latent space via a pre-trained Stable Diffusion VAE instead of pixel space.
- Achieved  $8.8 \times$  training speedup ( $4.05s \rightarrow 0.45s/iter$ ) through FP16 mixed precision, 8-bit Adam, and gradient checkpointing on a single RTX 3060 6GB.
- Reduced inference latency by  $6.1 \times$  ( $3.87s \rightarrow 0.63s/image$ ) by replacing the 200-step DDPM sampler with 50-step DDIM and compiling the UNet to a TensorRT engine with layer fusion and native FP16 execution.

### MedSAM CUDA-Accelerated Medical Segmentation | *PyTorch, CUDA, Vision Transformers*

- Implemented CUDA-accelerated PyTorch training pipelines to fine-tune and evaluate MedSAM on Kvasir-SEG.
- Fine-tuned ViT-based image encoder and mask decoder while freezing prompt encoder per MedSAM design.
- Improved Dice score from 0.908  $\rightarrow$  0.9448 through domain-specific fine-tuning on GPU setups.